

Bayesian Inference

A Graduate Introduction

Quantitative Methods Seminar

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Department of Statistics

Foundations

Prior Distributions

Conjugate Models

Computation: MCMC

Model Comparison

Foundations

The Bayesian Paradigm

Core Principle: Probability represents *degrees of belief*, not mere frequencies.

Bayes' Theorem

For parameters θ and data y :

$$\underbrace{p(\theta | y)}_{\text{posterior}} = \frac{\overbrace{p(y | \theta)}^{\text{likelihood}} \cdot \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(y)}_{\text{evidence}}}$$

Key Insight: We update our beliefs about θ after observing data y .

- **Prior** $p(\theta)$: Belief before seeing data
- **Likelihood** $p(y | \theta)$: How likely the data is, given θ

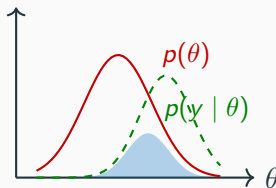
The Denominator: Evidence

The marginal likelihood (evidence) normalizes the posterior:

$$p(y) = \int p(y | \theta) p(\theta) d\theta$$

Interpretation:

- Average likelihood weighted by prior
- Often intractable $\Rightarrow$ MCMC methods
- Used for model comparison (Bayes factors)



Posterior Summary

Posterior Distribution

The posterior $p(\theta | y)$ is a *complete* summary of our knowledge about θ .

Common summaries:

- **Posterior mean:** $\mathbb{E}[\theta | y] = \int \theta p(\theta | y) d\theta$
- **Posterior mode (MAP):** $\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta | y)$
- **Credible intervals:** $\mathbb{P}(\theta \in [L, U] | y) = 1 - \alpha$

Credible vs. Confidence Intervals

- **Bayesian:** $\mathbb{P}(\theta \in CI | y) = 0.95$ (probability about parameter)
- **Frequentist:** $\mathbb{P}(CI \text{ contains } \theta) = 0.95$ (probability about procedure)

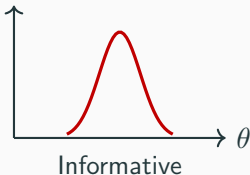
Prior Distributions

Choosing Priors

Priors encode domain knowledge or regularization.

Informative Priors

- Derived from previous studies
- Expert elicitation
- Regularization (shrinkage)



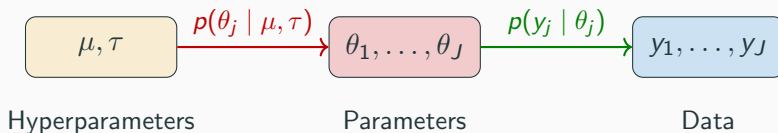
Non-informative Priors

- Uniform: $p(\theta) \propto 1$
- Jeffreys prior
- Weakly informative



Hierarchical Priors

When parameters themselves need priors, we build hierarchies:



Example: Multiple groups (schools, hospitals, cities) with shared but partially pooled estimates.

$$\theta_j \sim \mathcal{N}(\mu, \tau^2), \quad y_j \sim \mathcal{N}(\theta_j, \sigma_j^2)$$

Conjugate Models

Beta-Bernoulli Model

Setup: Observe y successes in n Bernoulli trials with success probability θ .

$$y_i \stackrel{iid}{\sim} \text{Bernoulli}(\theta), \quad i = 1, \dots, n$$
$$p(\theta) = \text{Beta}(\alpha, \beta), \quad \theta \in [0, 1]$$

Posterior Derivation

$$\begin{aligned} p(\theta \mid y) &\propto p(y \mid \theta) p(\theta) \\ &\propto \theta^{\sum y_i} (1 - \theta)^{n - \sum y_i} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1} \\ &= \theta^{\alpha + \sum y_i - 1} (1 - \theta)^{\beta + n - \sum y_i - 1} \end{aligned}$$

$$\theta \mid y \sim \text{Beta}(\alpha + \sum y_i, \beta + n - \sum y_i)$$

Beta Posterior Visualization

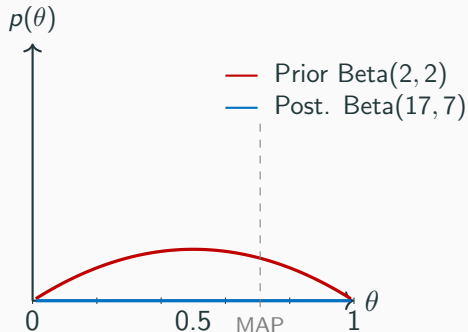
Example: $n = 20$ trials, $y = 15$ successes.

Prior: Beta(2, 2) (weakly informative)

Posterior: Beta(17, 7)

Posterior mean:

$$\mathbb{E}[\theta \mid y] = \frac{17}{17 + 7} = \frac{17}{24} \approx 0.708$$



Normal-Normal Conjugacy

Setup: Data from a normal distribution with known variance.

$$y_i \stackrel{iid}{\sim} \mathcal{N}(\theta, \sigma^2), \quad \sigma^2 \text{ known}$$
$$p(\theta) = \mathcal{N}(\mu_0, \tau_0^2)$$

Posterior Distribution

$$\theta \mid y \sim \mathcal{N}(\mu_n, \tau_n^2)$$

where:

$$\mu_n = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{y}}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}, \quad \frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}$$

Intuition:

Computation: MCMC

When Conjugacy Fails

For non-conjugate models, we cannot compute the posterior analytically:

$$p(\theta \mid y) = \frac{p(y \mid \theta) p(\theta)}{\int p(y \mid \theta') p(\theta') d\theta'} \quad (\text{intractable})$$

Solution: Draw samples $\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(S)}$ from $p(\theta \mid y)$.

Monte Carlo approximation:

$$\mathbb{E}[g(\theta) \mid y] = \int g(\theta) p(\theta \mid y) d\theta \approx \frac{1}{S} \sum_{s=1}^S g(\theta^{(s)})$$

Question: How do we generate samples from the posterior?

Metropolis-Hastings Algorithm

Algorithm

Initialize $\theta^{(0)}$. For $s = 1, 2, \dots, S$:

1. Propose: $\theta^* \sim q(\theta^* | \theta^{(s-1)})$
2. Compute acceptance probability:

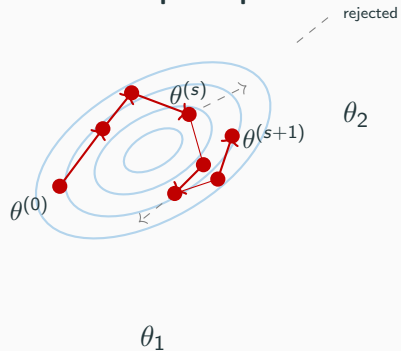
$$\alpha = \min \left(1, \frac{p(\theta^* | y) q(\theta^{(s-1)} | \theta^*)}{p(\theta^{(s-1)} | y) q(\theta^* | \theta^{(s-1)})} \right)$$

3. Accept with probability α : $\theta^{(s)} = \theta^*$; else $\theta^{(s)} = \theta^{(s-1)}$

Key Properties:

- Only need $p(y | \theta) p(\theta)$ up to proportionality (no evidence!)
- Under mild conditions, $\theta^{(s)} \xrightarrow{d} p(\theta | y)$ as $s \rightarrow \infty$

Random walk explores posterior



Gibbs Sampling

For multivariate $\theta = (\theta_1, \theta_2, \dots, \theta_p)$:

Gibbs Sampling Algorithm

Iterate through coordinates, sampling each from its full conditional:

$$\theta_1^{(s)} \sim p(\theta_1 \mid \theta_2^{(s-1)}, \dots, \theta_p^{(s-1)}, y)$$

$$\theta_2^{(s)} \sim p(\theta_2 \mid \theta_1^{(s)}, \theta_3^{(s-1)}, \dots, \theta_p^{(s-1)}, y)$$

$\vdots$

$$\theta_p^{(s)} \sim p(\theta_p \mid \theta_1^{(s)}, \dots, \theta_{p-1}^{(s)}, y)$$

Advantages:

- No rejection (always accept proposed sample)
- Single-parameter conditionals often tractable even when joint is not

Model Comparison

Bayesian Model Comparison

Given competing models $\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_K$:

Bayes Factor

The evidence ratio for models i and j :

$$\text{BF}_{ij} = \frac{p(y \mid \mathcal{M}_i)}{p(y \mid \mathcal{M}_j)} = \frac{\int p(y \mid \theta_i, \mathcal{M}_i) p(\theta_i \mid \mathcal{M}_i) d\theta_i}{\int p(y \mid \theta_j, \mathcal{M}_j) p(\theta_j \mid \mathcal{M}_j) d\theta_j}$$

Interpretation (Jeffreys' scale):

$\log_{10} \text{BF}_{ij}$	Strength of evidence
0 to 0.5	Not worth more than a bare mention
0.5 to 1	Substantial
1 to 2	Strong
> 2	Decisive

Deviance Information Criterion (DIC)

$$\text{DIC} = -2 \log p(y \mid \hat{\theta}_{\text{Bayes}}) + 2 p_D$$

where p_D is the effective number of parameters (penalizes complexity).

Widely Applicable Information Criterion (WAIC)

$$\text{WAIC} = -2 \sum_{i=1}^n \log \mathbb{E}_{\theta}[p(y_i \mid \theta)] + 2 \sum_{i=1}^n \text{Var}_{\theta}[\log p(y_i \mid \theta)]$$

Leave-One-Out Cross-Validation:

$$\text{LOO-CV} = \sum_{i=1}^n \log p(y_i \mid y_{-i}) \approx \sum_{i=1}^n \log \left(\frac{1}{S} \sum_{s=1}^S p(y_i \mid \theta^{(s)}) \right)$$

Summary: Key Takeaways

1. **Bayesian inference** treats parameters as random variables with distributions
2. **Posterior** $\propto$ Likelihood $\times$ Prior — update beliefs with data
3. **Conjugate priors** yield closed-form posteriors (Beta-Bernoulli, Normal-Normal)
4. **MCMC** (Metropolis-Hastings, Gibbs) enables inference for complex models
5. **Model comparison** via Bayes factors, DIC, WAIC, LOO-CV

References

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis*. CRC Press.